FINAL PROJECT

Treasure Adventure Story

Computer Graphics

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INTRODUCTION

1.1 Abstract

This project presents a 2D animated story using OpenGL
The story follows Sara and Mohammed on an adventure to find a hidden treasure map, travel through different places, solve obstacles, and finally discover the treasure

The project includes animations, transformations, textures, keyboard interaction, and multiple **scenes**

1.2 Introduction

The goal of this project is to apply computer graphics concepts using OpenGL and GLUT
The project was designed as an interactive story consisting of multiple scenes. Different graphical objects were drawn manually using geometric shapes such as circles, polygons, triangles, and lines

Several computer graphics concepts were applied, including:

Translation
Rotation
Scaling
Animation
Keyboard interaction
Texture mapping

The story contains five scenes:

Finding the treasure map
Walking through the forest
Discovering clues near the cave
Solving the rock obstacle
Finding the treasure

METHODOLOGY

2.1 IMPLEMENTATION DETAILS

The project was implemented using:
OpenGL
GLUT / FreeGLUT
C++
The program was divided into multiple functions to improve organization and readability

Examples of functions:
drawCharacter()
drawTree()
drawCloud()
drawTreasure()
drawCave()
Animation was implemented using timer functions and object movement updates

Keyboard interaction was added:
Press ‘N’ to move between scenes

Textures were added in:
Desert scene
Cave scene
Transformations such as translation, rotation, and scaling were used throughout the project

RESULTS

SCENE 1 – FINDING THE TREASURE MAP

Sara and Mohammed discover an old treasure map near their house
The scene includes a house, treasure box, and animated characters
The map appears after interaction with the mouse

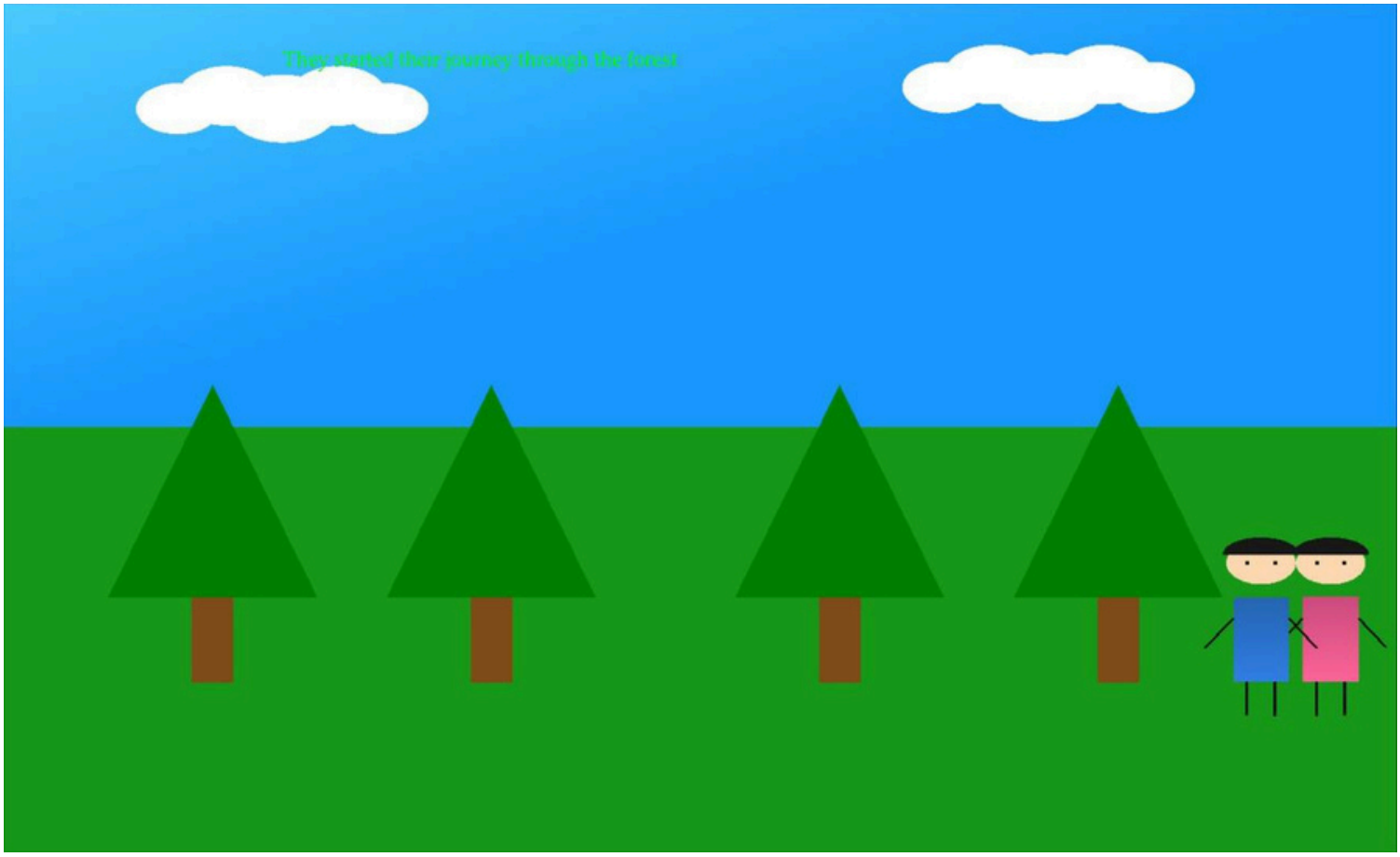


`Code Implementation:`
Different OpenGL primitives such as polygons, quads, and lines were used to build the objects in this scene

RESULTS

SCENE 2 – JOURNEY THROUGH THE FOREST

Sara and Mohammed begin their adventure through the forest
This scene contains animated movement, trees, clouds, and character walking animations

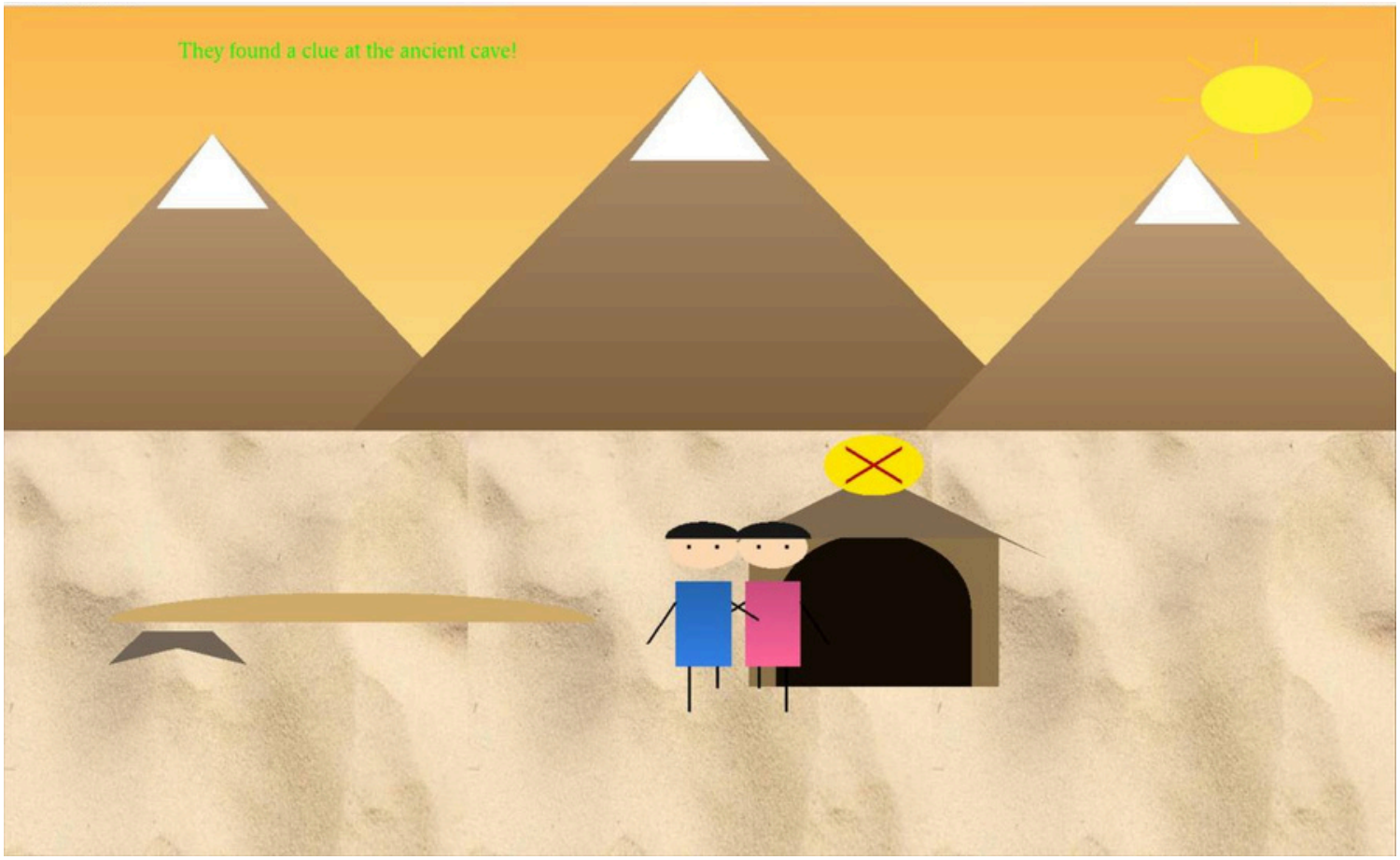


Code Implementation:
Different geometric shapes and color gradients were used to create the forest environment and sky background

RESULTS

SCENE 3 – DISCOVERING THE ANCIENT CAVE

The characters arrive at the mountains and discover an ancient cave
A clue appears near the cave to guide them to the next challenge
Texture mapping was used for the desert ground

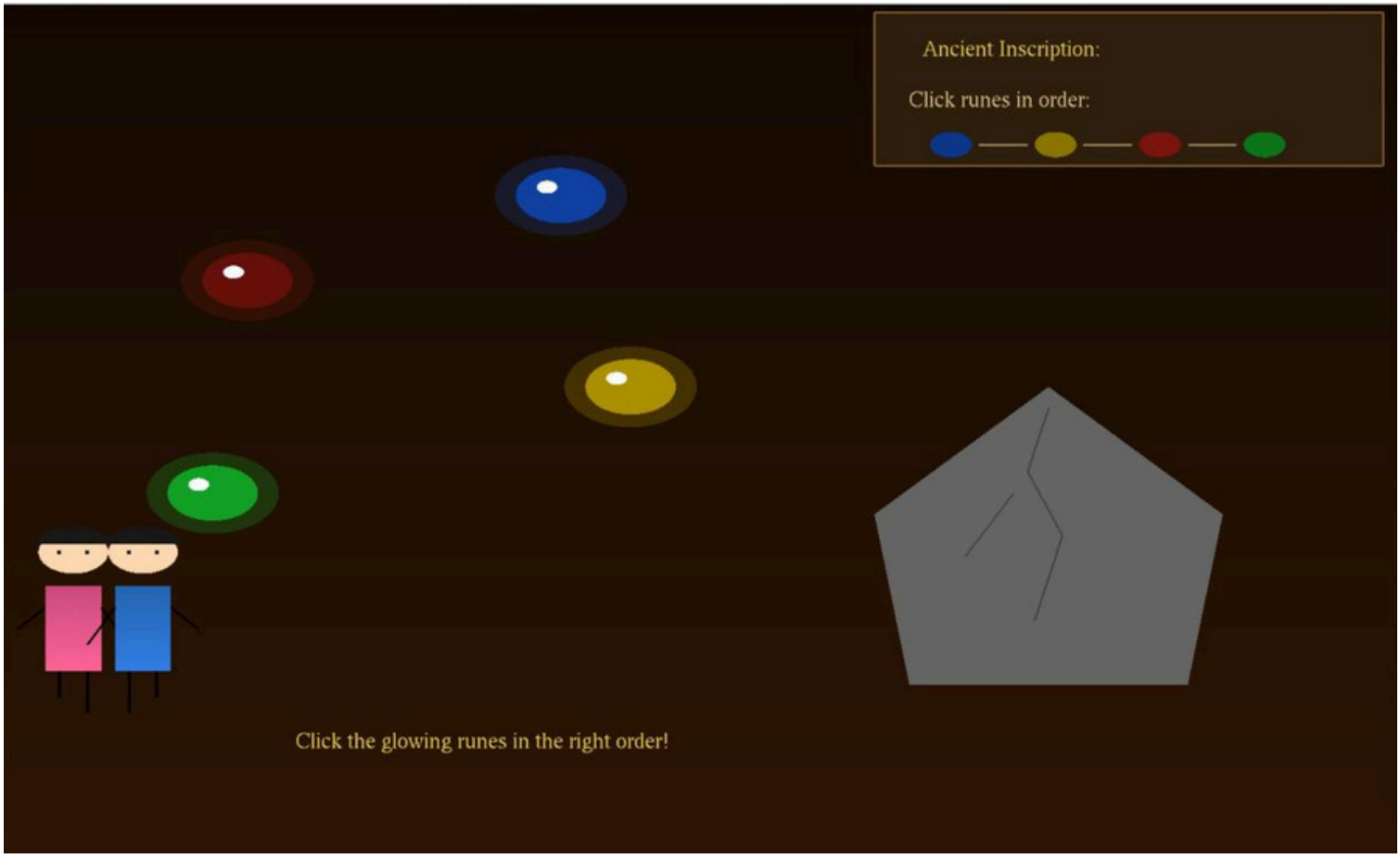


Code Implementation:
A sand texture image was applied to the ground using texture mapping techniques to create a more realistic desert environment

RESULTS

SCENE 4 – SOLVING THE RUNE PUZZLE

Inside the dark cave, the characters face a magical puzzle
The player must click the glowing runes in the correct order to continue the adventure
This scene includes interaction, glowing effects, and puzzle logic



`Code Implementation:`
Mouse input and conditional statements were used to check the correct rune sequence and control the puzzle progress

RESULTS

SCENE 5 – FINDING THE TREASURE

After solving the puzzle, Sara and Mohammed finally find the treasure chest inside the cave
The final scene includes cave textures, treasure objects, and celebration elements



Code Implementation:
A cave background texture image was added to create a more realistic final environment Glow effects and decorative treasure elements were used to make the ending scene more visually appealing

CONCLUSION & FUTURE WORK

In this project, we learned how to build a complete graphical story using OpenGL
We applied many computer graphics concepts such as animation, transformations, textures, and user interaction

Future improvements may include:
Adding sound effects
Creating smoother animations
Adding more advanced textures
Using 3D graphics instead of 2D

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REFERENCES

OpenGL Documentation
FreeGLUT Documentation
Microsoft Visual Studio
OpenGL Tutorials